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The Incremental Validity of a Short Form of the Ideational Behavior Scale and Usefulness of Distractor, Contraindicative, and Lie Scales

ABSTRACT

This article describes an empirical refinement of the *Runco Ideational Behavior Scale* (RIBS). The RIBS seems to be associated with divergent thinking, and the potential for creative thinking, but it was possible that its validity could be improved. With this in mind, three new scales were developed and the unique benefit (or "incremental validity") of each was assessed. The three validity scales contained (a) distractor items, (b) contraindicative items, or (c) items constituting a lie scale. Step-wise analyses using the three validity scales indicated very little incremental validity: They were interpreted in the light of psychometric theory, which suggests that their contribution may have been on a behavioral rather than on a statistical level. One additional analysis explored the possibility that a short form could be constructed. The short form of the RIBS was highly correlated with the long form (r = .94). Thus, most decisions made by the short form will be entirely compatible with decisions that would have been reached using the long form. A second significant result was that the RIBS was highly correlated with a check list of creative activities, supporting its concurrent validity.

Keywords: ideation, divergent thinking, originality, incremental validity, concurrent validity, validity scales.

Psychometric research plays a vital yet largely unsung role in studies of creativity. In fact, it plays two roles: First it insures that empirical research relying on any sort of assessment is valid and therefore meaningful. No inferences about the meaning of creativity in the natural environment or predictions about future creative accomplishments would be possible without psychometrics. The second role of psychometric theory is its contribution to accurate diagnoses and identification of individuals

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with creative potential. This is in turn related to the accurate delineation of specializations (e.g., particular domains). Essentially, without psychometrics, no generalizations of any sort from empirical findings would be warranted.

The present psychometric study examines one kind of the validity, and in particular the *incremental validity,* of the criterion measure known as the *Runco Ideational Behavior Scale*. Several investigations have demonstrated the usefulness of the RIBS as a criterion measure in studies of divergent thinking and creative ideation (Plucker, Runco, & Lim, 2006; Runco, Plucker, & Lim, 2000–2001). Yet there is room for improvement. The present investigation used the concept of incremental validity to identify exactly how to improve the RIBS.

CREATIVE IDEATION

Runco (2007) suggested that the word creativity be avoided because of its ambiguity and because of the arguments surrounding definitions. His idea was to always use the adjective "creative" instead of the noun "creativity." This would require that every discussion of creativity is concrete and specific. Instead of sweeping and questionable claims about "creativity," discussions would point specifically to "creative accomplishment," "creative potential," "creative product," and so on.

This line of reasoning is relevant here because the focus is on ideational behaviors. Ideation, which is the label given to the process resulting in ideas, is probably involved in all creative activity. Certainly, there is more to creative activity than the having of ideas, and ideas are not always creative. Still, ideas may be involved in all creative efforts, at least early on. That is one reason why tests of divergent thinking (Guilford, 1968; Runco, 1991; Torrance, 1995) are so often used to assess creative potential. Tests of divergent thinking are open-ended and allow an examinee to produce a large number of ideas. That is the difference between tests of divergent and convergent thinking: The former allows the examinee to construct new and original ideas, the latter focuses on information drawn from long-term memory and experience.

Runco (2013) argued that ideas can be treated as products of original, divergent, and creative thinking. Ideas are also less likely than achievements and accomplishments to be biased by opportunity, chance, and socioeconomic status. Note, however, that ideas probably play a role in everyday as well as eminent creativity. They are involved in all domains, including the everyday domain of creative effort. The ideation captured by the RIBS is thus fair and general, characteristic of everyone and all domains of performance. The items in the RIBS (examples given in the Method section, below) describe observable behaviors that depend on ideation. Individuals who score high on the RIBS frequently generate and appreciate ideas. They frequently think about alternate ways of solving problems and getting things done.

The RIBS was originally developed as a criterion measure. Too often, tests of divergent thinking have been validated inappropriately. Criterion measures of creative accomplishment or achievement have been used, for example, even though divergent thinking tests are intended to assess ideation and the potential for creative problem solving, not past accomplishment. To make matters worse, domain-specific criteria are often used. These ask about achievement in art, science, leadership, music, or the like, even though those surely require domain-specific skills, knowledge, and practice, and not just ideation. No wonder predictive validities in studies of divergent thinking tended to be moderate or low (Hocevar, 1980; Runco, 1991, 2013). Again, the criteria in past research have often been quite inappropriate. The RIBS was developed to rectify this and to obtain more accurate predictive validities for tests of divergent thinking. Because divergent thinking tests assess ideation, a scale emphasizing ideation must be used as the criterion.

PREVIOUS RESEARCH ON THE RIBS

Runco et al. (2000–2001) reported that RIBS scores were statistically unrelated to two scales from a measure of creative attitudes and, perhaps most importantly, uncorrelated with GPA. The last of these is sometimes used as an estimate of traditional intellectual ability, and although not synonymous with the IQ or "g," the results from Runco et al. (2000–2001) clearly support the discriminant validity of the RIBS. Exploratory and confirmatory factor analyses with two different samples were not as clear. There was a hint of a two-factor solution, which is at odds with the design as a one-factor instrument.

Plucker et al. (2006) used the RIBS in an empirical test of *the two-tiered theory* of creativity (Runco & Chand, 1995). This theory describes the creative process in terms of problem finding, ideation, and evaluative thinking (on the primary tier) and motivation (intrinsic and extrinsic) and knowledge (declarative and procedural) on the secondary tier. Plucker et al. were especially interested in the impact of motivation on the creative process and the interaction between motivation and divergent thinking (DT). They administered the RIBS as a criterion and tests of divergent thinking to capture the ideational component of the process. Motivation was estimated with a novel methodology, which provided data on *discretionary time on task* (DToT). This was defined as the amount of time that an examinee chose to invest in his or her work. It was just the amount of time (in seconds) and was discretionary because each person was allowed to choose when to stop working. The DT battery included two realistic problems, which required problem generation (PG), and two Instances tests (i.e., "name all of the things you can think of that make noise" and "list all of the things you can think of that move on wheels"). These provided both fluency and originality scores. Plucker et al. reported that DT test scores predicted ideational behavior (RIBS scale), which confirms the usefulness of the RIBS as a criterion for studies on original and divergent thinking. Interestingly, there was no indication of any curvilinear trends. These were tested because it is reasonable to assume that DT and DToT might be related to ideational behavior, but only up to a point.

Clapham, Muchlinski, and Sedlacek (2005) were fairly critical of the RIBS. They administered it along with the Figural scale from the *Torrance Test of Creative Thinking* and the *How Do You Think Test* (Davis, 1975). The correlation between the RIBS and TTCT was statistically significant (.05), but far from sizable. The two shared less than 10% of their variance. The RIBS was the only measure that was related to the other two estimates of creative potential. If all three measures were assessing creative potential, they should be related, but, for some reason, the TTCT and the *How Do You Think* were unrelated to one another.

This last result may have resulted from an irregularity with the test administration. In fact, there was a critical misunderstanding in this investigation. Clapham et al. claimed that the RIBS and the TTCT "measured the same construct." This particular misunderstanding is not uncommon in the creativity literature. Indeed, it was quite common 10–20 years ago to view tests of divergent thinking, such as the TTCT, the divergent production tests developed by Guilford (1968), or the battery administered by Wallach and Kogan (1965), as tests of creativity. This essentially equates tests of divergent thinking and creativity, which makes no sense. As is the case with all tests, measures of DT provide scores that are based on samples of behavior. They do not measure actual creative performances. Yet, they were often treated in that fashion. To make matters worse, DT tests were sometimes used as criteria of creative talent, when in fact they are predictors (Wallach, 1970). There is a huge difference between a predictor and a criterion, and an equally large difference between the measurement of potential and the assessment of actual creative performance. Tests of divergent thinking are "estimates of the potential for creative thinking" (Runco, 1999, p. 581). The RIBS, on the other hand, is a criterion measure. Clapham et al. seemed to have overlooked all of this. At one point, they referred to the RIBS as a biographical inventory, and later referred to a non-focused ideation scale of the RIBS, which is contrary to its actual design. There were, then, several misunderstandings that may help explain why their results are the only ones to date that do not offer clear support for the RIBS as a criterion measure.

One part of psychometric theory may not be obvious when interpreting the result summarized above. That is the fact that, in psychometric theory, tests of statistical significance are not all-important. In other areas of empirical research, social and behavioral scientists rely on tests of significance, and usually on the .05 and .01 probability levels. In psychometric theory, in contrast, validity and reliability are treated as continuous indicators, and different levels of agreement (i.e., correlation coefficients) are acceptable, depending on the particular test and application. In fact, sometimes, quite low correlations coefficients are adequate, perhaps because the test is used as one of several sources of information, or because the instrument targets something that is known to be quite subjective. Projective tests may best exemplify this last point.

A number of other investigations have supported the use of the RIBS (Pannells & Claxton, 2008; Ames & Runco, 2005; Vincent, Decker, & Mumford, 2002; Walczyk, Runco, Tripp, & Smith, 2008). Pannells and Claxton (2008), for example, found that RIBS scores were associated with external, but not internal locus of control. Walczyk et al. (2008) found RIBS scores to be predictive of lies given within special DT tests. These had been constructed specifically to identify lying. The interesting thing was that here the RIBS was more accurate than scores from DT tests. Ames and Runco (2005) found much the same in their investigation of entrepreneurs. In both of these last two investigations, the RIBS seemed to be more sensitive to individual differences than tests of DT.

INCREMENTAL VALIDITY

The RIBS seems to be a useful criterion for DT tests or other methods used to study ideation. Yet, it was probable that an examination of the *incremental validity* (Anastasi, 1982) of the RIBS could make it that much more accurate. Incremental validity can be defined in terms of improvements in validity resulting from the addition of predictors, scales, or indicators. Three new scales were added to the RIBS (which made it the RIBS-V) and their impact was empirically determined. The usefulness of a short form of the RIBS (the RIBS-S) was also examined. The typical rationale and benefits of short forms of tests justified its construction. Short forms are often useful in research and education because they can be given in a brief period of time, preclude fatigue, and do not disrupt regular activities to the same degree as a long test. Admittedly, psychometric theory holds that in general, longer assessments will be more accurate than briefer ones (Nunnally, 1978). Longer tests examine larger samples of behavior, so results tend to be more representative of what is typical. Also, errors (which are assumed to be random) tend to cancel themselves out if a test is long enough for them to do so, and as a result, reliability is high. Yet, a short form can be adequately reliable and valid. The present study examined both the long form of the RIBS-V and the new short form and compared their respective reliability and validity.

The short- and long forms of the RIBS were examined using a regression approach. This allowed a clear comparison of the two forms, but also allowed specific possible improvements to the RIBS to be tested as well. Improvements were expected to result from the inclusion of the three new scales, which we refer to as “validity scales” as they were justified solely by the interest in improving the validity of the RIBS. The unique contribution of each new scale was determined in separate steps of the regression analyses. In sum, as is apparent in the Method and Results, instead of merely comparing a short form with the original (long) form of the RIBS, analyses were conducted to compare the validity of the short form (a correlation between it and various other criteria) with (a) that of the long form, and (b) that provided by each validity scale, added one by one to the regression equation.

The validity scales tested here represented a *distractor scale*, a *lie scale*, and a scale containing *contraindicative* items. The distractor scale items asked respondents about things other than ideation. Distractor scales are used so examinees cannot infer the intent of a test. If they infer the intent, they can more easily manipulate the scores. Distractor scales keep a test from being transparent and thereby minimize the possibility that the test score only indicates how the examinee wants to be perceived.

Contraindicative items are also used in many instruments (e.g., on the *Adjective Check List* [Gough, 1980] and the *Teachers’ Evaluation of Students’ Creativity* [Runco, 1984]) and are quite useful for minimizing measurement error resulting from response sets. A response set implies that the respondent may not be reading the items carefully and is instead responding based on some pattern of response options. A simple example would be someone who, on a true–false test, merely gives True, then False, then True, then False, then True, then False, and so on. On a multiple choice test, the person may answer A, then B, then C, then D, and then A, then B, and so on. Clearly, this kind of responding is not indicative of ability or anything about the examinee.

The lie scale was suggested by one of the validity scales of the MMPI. The basic idea is that accuracy of responding can be determined by asking about things nearly everyone would deny or reject. Very few people will agree with a question such as "I never wonder what other people are thinking," for example, or "I never do something and later regret it." If an examinee does agree every question of this sort, he or she may tend toward fabrication or responses that are not truthful. The RIBS–V included 12 such items. A high score on the scale calculated from these items would bring all responses on the RIBS into question.

In sum, the objectives of this investigation were to use a regression approach to the following ends:

1 Determine the usefulness of a short form of the RIBS and check its correlation with the long form of the RIBS.
2 Empirically assess the incremental benefits (i.e., impact on relevant psychometric coefficients) of the three aforementioned validity scales, included in the RIBS–V, as possible improvements on its reliability and validity.

METHOD

PARTICIPANTS

A total of 152 undergraduate and graduate students, enrolled in psychology courses at a University in the Southern United States, were offered extra credit in exchange for their participation in this investigation. The mean age of the sample was 23.41 (SD = 5.23). The racial composition of the participants was diverse: European American-107; African American-35; Hispanic American-4; Asian American-3; Native American-3. There were several more women (N = 85) than men (N = 67) in the sample.

PROCEDURE

All participants were treated in accordance with the ethical guidelines set forth by the American Psychological Association (APA, 2010). Prior to data collection, approval was obtained from the IRB at the institution sponsoring this research. Participants received the instruments and a demographic sheet in the form of take-home booklets, which they completed at their convenience and returned to their instructors. The order of instruments was randomized across individuals to control for possible order effects. Participants were not made aware of the hypotheses until after all the survey booklets were completed and returned. To protect the participants' anonymity, numbers (not names) were used to label test booklets. In addition, all participants signed separate, detachable consent forms that indicated that all test data would be kept anonymous.

INSTRUMENTS

Runco ideational behavior scale

This version of the RIBS contained 74 items. Within those items were 19 that were eventually used to generate a score for the short form, 12 in the lie scale, 7 contraindicative items, and 13 distractor items. Recall here that the lie scale contains items that would probably have only very infrequent agreement by a respondent (e.g., "I rarely think about what other people are thinking," "I never think back on what I have done and consider what else I might have done instead," "I do not think about what it would be like to make more money"). The contraindicative items are merely negatively worded so that a high score indicates low ideation (e.g., "I avoid an activity which requires on-the-spot problem solving" or "I have thoughts which can block out all other thoughts—it is like I am stuck in a rut"). The distractor items insure that the intent of the RIBS is not transparent; they have little, if anything, to do with ideation (e.g., "I am reflective").

The 19 items with the short form were selected *a priori* because they are explicitly and unambiguously tied to ideational behavior (e.g., "I have ideas about what I will be doing in the future," or "I have trouble sleeping at night, so many ideas are bouncing around in my head"). They do not imply much if anything about attitude, interest, or any other correlate of ideation; they are pure in the sense that they only involve the having of ideas.

Something should be said about the format of the RIBS used in this investigation. That is because, in addition to testing a short form (RIBS-S) and examining the incremental variance contributed by the validity scales, one further yet minor innovation in the RIBS was introduced. It was not something that led to another check of incremental validity, but was merely a new more logical format for the response options. Unlike the previous versions with a typical 1-2-3-4-5 Likert scale and conventional verbal anchors (e.g., "always", "sometimes", "never"), the version used here had 0-1-2-3-4 as the numeric options. This retained the five options, but was more sensical because one option was labeled, "never", which is well-matched with zero rather than 1. The entire list of options, given for each RIBS question, was as follows: 0 (never), 1 (approximately once a year), 2 (once or twice each month, approximately), 3 (once or twice each week, approximately), and 4 (just about every day, sometimes more than once each day). These were recoded for the items in the contraindicative scale before composites were formed and items averaged. This recoding merely changed 0 to 4, 1 to 3, 3 to 1, and 4 to 0.

The verbatim directions for the RIBS were:

> "Use the 0–4 scale (given below) to indicate how often each of the phrases describes your thinking. Note the focus on your thinking, which might be different from your actual behavior. Also, you may need to approximate. Please indicate how you really think, not how you believe you should act. Remember—no names are used. Your responses are confidential."

They were also given a reminder midway through the RIBS: "*Again, you may need to approximate. For each item, circle the response option that is THE CLOSEST to being accurate.*"

THE CREATIVITY ASSESSMENT PACKAGE

Although this was not an investigation of predictive validity, additional measures were administered and used as criteria for the analyses of incremental validity. The first was the *Creativity Assessment Package* (CAP; Williams, 1980). It was developed as a test of creative potential and provides fluency, flexibility, originality, elaboration, and titles scores. It provides examinees with frames in which they can draw. Fluency, then, is defined in terms of the number of frames used by an examinee. Originality is defined in terms of how often an examinee draws things outside a frame. The CAP is therefore quite different from DT.

THE CREATIVE ACTIVITY AND ACHIEVEMENT CHECKLIST (CAAC)

The second criterion for the regression analyses was a 45-item *Creative Activity and Achievement Checklist* (CAAC) similar to those used by Holland (1961), Hocevar (1980), Milgram and Milgram (1978), Runco (1986), and Wallach and Kogan (1965). This version sampled five domains: Science, writing, art, crafts, and music. Questions asked "how often have you…" and then named a particular activity (e.g., "written a poem"). Responses were Likert type with the conventional five options (never, rarely, sometimes, often, always) mentioned just above. Okuda, Runco, and Berger (1991) reported good inter-item reliabilities for the various domains (.71 < *r*s < .91).

RESULTS AND DISCUSSION

RATIONALE FOR ANALYSES

Sechrest (1963) recommended providing the evidence of incremental validity in order to show that the target test (in this case, the RIBS-S) produces better predictions than those already available. He argued that the multiple or partial *R*s are expected to increase when the target test is added, and in that case, corrections for attenuation should be made to examine if the increase is because of the reliability of measurement of the predictor variable. This increase could be indicated through the multiple regression approach, which allows testing the contribution of each test added at different steps to explain the total variance. Hunsley and Meyer (2003) pointed specifically to hierarchical regression analyses for examinations of incremental validity. Quite a few previous studies have followed this advice and, like the present analyses, employed multiple regression (e.g., Clevenger, Pereira, Wiechmann, Schmitt, & Harvey, 2001; Day & Silverman, 1989; Watkins & Glutting, 2000).

RELIABILITY AND INTERCORRELATIONS

The first analysis was simply a product moment correlation between the long form and the short form ($r = .94$, $p < .001$). This confirmed that any decisions using information from the short form are likely to correspond very closely to decisions made based on the long form. In the terms used earlier when incremental validity was defined (Anastasi, 1982), the number of errors of identification is unlikely to increase significantly if the short form is used instead of the long form. If someone with frequent ideation is identified with the long form, they will nearly always also be correctly identified with the short form. Table 1 presents the means and standard deviations for the RIBS-V and RIBS-S.

Reliability coefficients (alphas) are also presented in Table 1. These are useful as indices of the internal consistency of the RIBS, but they are also useful because all correlational analyses, including the product moment analysis reported above and the regression analyses reported below, are influenced by the reliability of the measures involved. This may be easiest to understand by keeping in mind that (a) a reliability coefficient is an estimate of how much variance can be explained, and (b) that portion of the variability which cannot be explained is error variance or measurement error. Correlational analyses cannot explain error; the upper limit for any validity coefficient is, then, equal to the magnitude of a reliability coefficient. In fact, if the coefficient of agreement above (.94) was adjusted to take the attenuation due to reliability into account, the result would exceed 1.00.

The alpha for the contraindicative scale was quite low. It may have been that examinees were confused by the reversal in the direction of the questions (they were accustomed to giving a response such as "daily" but then had to jump to the opposite end of the scale to response such as "never"), but as noted above, contraindicative items are used in various other measures, and reversals such as these should insure a careful reading of the RIBS items by the examinees.

CONCURRENT VALIDITY

The validity of the RIBS was examined by correlating each form of it with two criterion measures. One of these, the CAP, had provided fluency, flexibility, titles, elaboration, and originality scores. Each of these five scores was based on 24 items. One item was omitted from the flexibility scale because it lowered alpha, and alpha

TABLE 1. Descriptive and Inter–item Reliability of RIBS and Validity Scales (N = 152)

	M	*SD*	**Alpha**
RIBS			
Long (RIBS-V)	100.05	23.81	0.93
Short (RIBS-S)	42.43	11.05	0.84
Alternative Form	57.9	13.84	0.90
Validity Scales			
Lie	26.05	7.52	0.83
Distractor	30.05	8.04	0.84
Counterindicative	13.88	3.98	0.46

Note. Alpha (Cronbach's) was used for inter-item reliability.

was already questionable (alpha = .60). The other coefficients were .93 (Fluency), .95 (Titles), .82 (Elaboration), and .87 (Originality). The other measure used to estimate validity, the CAAC, was also reliable, with alphas of .84, .79, .76, .65, and .80 for science, writing, art, crafts, and music, respectively. A total CAAS composite was found by averaging all CAAS items (alpha = .87). A composite score was generated for each domain by averaging the items within each.

Regression analyses used either the CAP scores or the CAAS scores as criterion and RIBS as predictor. The most important analysis, given our objectives, may have been the regression that used the CAAS composite as the dependent measure (or criterion) and then entered the predictors in steps, first the short form, then the long form, and finally, in one step, the distractor, contraindicative, and lie scales from the RIBS-V. The second step was critical because it indicated the incremental variance or benefit of using the long form instead of just the short form for predictions of creative activity and accomplishment.

Results indicated that the short form (step 1) was significantly related to the CAAC ($r = .47$, $p < .001$). The other two steps in the regression indicated that the amount of added variance was not statistically significant ($R^2 = .010$ and .024, respectively, ns). Thus, the long form only added 1% of the variance to the prediction of the short form (which by itself accounted for 22%). That 1% was not statistically significant in the traditionally sense of alpha = .05 or .01 and is also insignificant in the psychometric sense. Recall here that in psychometric theory, tests of significance are not all-important. Instead, validity and reliability coefficients are interpreted much like effect sizes. This is why we started by introducing the concept of incremental validity and the idea that the more variance explained by validity coefficients, the more accurate the decisions that can be made by looking at test scores (Anastasi, 1982; Nunnally, 1978). Still, the long form only contributed 1% more variance, beyond the short form, and the addition of the lie, distractor, and contraindicative scales only another 2%.

Keeping in mind that tests of significance are secondary in psychometric studies, it is notable that the weights given to the lie scale in the third step of the regression, although not statistically significant, were negative. This is what would be expected from psychometric theory, and although the amount of variance explained by it was not statistically significant ($B = -.226$, $SE = -.403$), it may have been that the benefit of the inclusion of the three validity scales was that they insure that the responses to all RIBS questions are more thoughtful. In this light, if they do have an impact on the RIBS, it might be not in their boosting predictive or concurrent validity, but in minimizing response sets and increasing mindful responding. It should also be remembered that the contraindicative items had been recoded or they too would have had a negative weight ($B = .181$, $SE = .248$).

Interestingly, similar regression analyses with the CAP failed to uncover a relationship with any RIBS scale. Less than 10% of the RIBS variance was associated with the CAP. This may have reflected the unique format and focus of the CAP. Recall here that it relied on graphic frames and free drawing. It should not be confused with a test of DT.

PRODUCT MOMENT CORRELATIONS

Table 2 presents product moment coefficients to assist in the interpretation of the results. Note that the short form was less related to the lie scale than was the long form, but not by much (*r*s = .59 and .53, both *p*s < .001). Also noteworthy is the correlation between the short form and what might be considered to be an "alternative form" of the RIBS (i.e., the items not in the short form, but also not in any of the validity scales) was .81 (p = .001). There might in the future be some use for an alternative form of the RIBS (e.g., in studies of pre-post-treatment changes).

CONCLUSIONS

These results suggest one very simple conclusion: The short form of the RIBS (the RIBS-S) is probably adequate for the assessment of ideational behavior. This conclusion is based on the reliability of the RIBS-S, the bivariate correlation between the short form and long form, and the multiple regression analysis showing that the RIBS-S accounted for statistically significant portion of the variance in CAAC scores. The last of these is a validity coefficient, and not a bad one relative to other validity coefficients found in the creativity literature.

If there is any need to label the coefficient between RIBS and CAAC more precisely than merely "validity," it is probably best to call it *concurrent validity*. *Predictive validity* would be inappropriate. In statistical analyses, there is little difference between analyses of predictive validity and analyses of the sort done here. Both involve correlating one measure with another. The difference is mostly temporal: An investigation of predictive validity assumes that the criterion data are collected after the predictor data. This can be complicated when a measure is actually postdictive (asking about the past, as in the CAAC) because surely predictions look forward rather than backward! All in all, it seems best to view the analyses using the RIBS and CAAC as examinations of concurrent validity.

As emphasized above, validity coefficients (and reliability coefficients, for that matter) should not be evaluated by statistical significance (Anastasi, 1982; Nunnally,

TABLE 2. Intercorrelations Between Subscales of RIBS (N = 142)

	1	2	3	4	5	6
RIBS						
1. Short (RIBS-S)		.94*	−.09	.75*	.53*	.81*
2. Long (RIBS-V)			−.10	.84*	.59*	.96*
3. Counterindicative				−.08	−.30	−.10
4. Distractor					.57*	.85*
5. Lie						.58*
6. Alternative Form						

Note. Short Form (RIBS-S), Long Form RIBS (RIBS-V), three validity scales (Contraindicative, Distractor, and Lie Scale), and Alternate Form of RIBS.
*$p < .01$.

1978), but are more like effect sizes; some are inadequate, some adequate but not impressive, and some impressive. Even this suggests a trichotomy when in fact validity coefficients are best viewed as representing a continuous scale. More is of course better (i.e., more accurate predictions or, in the case of reliability, more agreement and consistency), but there are no absolute cut-off levels or tests of probability. We did report probability levels above, but interpretations should certainly not rely on them.

The correlational results might imply that validity scales are not necessary when the RIBS is administered. Recall, however, that although the three validity scales used in the RIBS-V might not have contributed much variance, they may have been useful on a behavioral level, keeping respondents on their toes and mindful. It certainly would not hurt to conduct research in the future to examine other validity scales, or improve somehow on those examined in the present research. The use of validity scales might be contrary to the rationale for a short form of the RIBS, but if a long form can be given, validity scales might indeed contribute to valid responding by examinees. For most purposes, it appears that accurate decisions will be made using the short form of the RIBS.

Putting the results of this study into the larger literature, we would suggest that there is reason to be optimistic about valid assessments of creativity. Still, care must be taken because what is being assessed by the RIBS, and by tests of divergent thinking, is the potential for creative thinking. This is very different from actual creative achievement or accomplishment. In fact, it may be that this is the most important direction for future research in the creativity literature: to discover how creative potential, as estimated by the RIBS and tests of divergent thinking, can be fulfilled and translated into actual creative accomplishment. For now, it is safe to conclude that there are measures available to conduct that type of research.

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